

MODEL QUESTION PAPER

TED(15) 4051

Revision 2015

FOURTH SEMESTER DIPLOMA EXAMINATION IN AUTOMOBILE ENGINEERING

APPLIED THERMODYNAMICS

Max.marks 100

Time 3 Hrs.

PART A

(Maximum Marks 10)

I Answer the following questions in one or two sentences.

1. Define a thermodynamic cycle
2. State Zeroth law of thermodynamics
3. Define air standard efficiency of a cycle
- 4 What is volume of free air delivered by a compressor?
5. State Fourier's law of thermal conduction. (5x2=10)

PART B

(Max. marks 30)

II Answer any five full questions

1. Explain any four thermodynamic properties in sixty words
2. Derive an expression for the work done and heat transferred during an adiabatic process
3. Compare the thermal efficiencies of Otto and diesel cycle
4. Explain the indicated mean effective pressure (imep) of an IC engine in sixty words
5. Explain the concept of black body, grey body and white body in sixty words
6. Illustrate the working of a single stage, single acting compressor with a sketch
7. Explain the Stefan-Boltzman law in sixty words. (5x6=30)

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PART C

(Max.marks 60)

(Answer any one full question from each unit)

UNIT I

- IIIa. Derive an expression for the work done and heat transferred during an isothermal Process (6)
- b. A gas having a volume of 0.5m^3 at 25°C and 210kPa is compressed adiabatically to 640kPa . Find its characteristic gas constant and molecular mass. Assume R_u as 8314 kJ/Kg mole K . (9)

OR

- IVa. Describe the following terms of thermodynamics in sixty words
i) system ii) state iii) process (6)
- b. Air at 60kPa and 25°C is expanded adiabatically to 15kPa . Calculate the work done and the change in internal energy. If the same expansion is isothermal in nature find the corresponding values (9)

UNIT II

- Va. Explain in sixty words the Otto cycle with P-V and T-S diagram (6)
- b. What is the thermal efficiency and compression ratio of an engine working on Otto cycle, working in a temperature range of 373K and 673K . Assume the value of adiabatic index as 1.41. (9)

OR

- VIa. Derive an expression for the air standard efficiency of an Otto cycle (6)
- b. A Diesel engine has a compression ratio of 16 and cut-off takes place at 6% of the stroke. Find air standard efficiency. Take adiabatic index as 1.4 (9)

VIIa. List six uses of compressed air (6)

b. Following observations were recorded during a test on four stroke diesel engine

Area of indicator diagram	0.0006m ²
Length of indicator diagram	0.04m
Spring number	40000kPa/m
Diameter of piston	180mm
Length of stroke	220mm
Engine speed	600rpm

Determine indicated mean effective pressure and indicated power. (9)

OR

VIIIa. Explain the following terms associated with air compressor

i) swept volume ii) capacity of the compressor ii) mean effective pressure (6)

b. A four cylinder four stroke cycle petrol engine has a stroke volume of 3.5 liters. The mean effective pressure exerted is 750kPa and crank runs at 600rpm. If there is an average misfire

of 3 per minute, find the indicated power of the engine. (9)

UNIT IV

IXa. Explain the classifications of heat exchangers with simple sketches in sixty words (6)

b. A copper rod 170mm long and of $7.85 \times 10^{-5} \text{m}^2$ area of cross section, thermally insulated, is heated at one end through 120°C while the other end is kept at 30°C . Calculate the amount of heat flow in 5 minutes. Thermal conductivity for copper is 380W/m K (9)

OR

Xa. Define the following terms i) absorptivity ii) reflectivity iii) transmissivity (6)

b. A brick wall 200mm thick is faced with concrete 30mm thick. If the temperature of the exposed brick surface is 50°C and that of the concrete is 5°C . Determine the heat loss per hour through the wall of 12m long and 3m height. Determine also the interface

temperature, given the thermal conductivities of the brick and concrete are $0.69 \text{W/m}^\circ\text{C}$ and $0.93 \text{W/m}^\circ\text{C}$ respectively (9)

